When Operator Accuracy Does Not Transfer to the Image: A Learned Interior Transport Operator for Faceted Gemstones

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Honours Research Report

***Abstract*—Neural appearance models can replace detailed light transport with a compact learned representation. This study investigates whether an adaptation of that approach reproduces the appearance of resolved, specular gemstones. A direction-conditioned radiance distribution map is first evaluated and then generalised to an interior transport operator conditioned on entry position, direction, surface normal and wavelength. The operator predicts finite-depth escape, an exit triangle, and an exit position and direction. Stochastic and deterministic decoders are evaluated on three procedural diamond cuts using spectral reference renders, independent render seeds, and a separation of operator-mediated light from untouched surface reflections. Learned exit-triangle prediction exceeds the tested lookup baselines, but this improvement does not ensure image agreement. At matched rendering budgets, stochastic decoders achieve pixel correlations of 0.136–0.293, while deep deterministic decoders achieve 0.038–0.168 despite improved validation triangle accuracy. Facet-averaged measurements reveal that coarse appearance can nevertheless be recovered: on the step cut, correlation reaches 0.899 while fine-scale agreement remains low. Recorded ablations implicate both branch selection and within-branch placement; they do not establish a universal limit on neural transport. The results support a narrower conclusion: under the tested studio illumination, operator-level accuracy and perceived sharpness are insufficient proxies for rendered fidelity. Training convergence, illumination coverage and computational cost are discussed with explicit separation of completed measurements from further validation.**

***Index Terms*—neural rendering, spectral light transport, gemstones, mixture density networks, image evaluation, Monte Carlo rendering**

I. Introduction

The appearance of a faceted diamond depends on where internally reflected and refracted light emerges. Small changes in entry position or direction can change the intersected facets and redirect a path. Narrow sources produce high-contrast structure sensitive to these changes; dispersion adds wavelength dependence. An approximation must therefore preserve both transmitted energy and its placement.

Soh and Montazeri [1] aggregate explicit cloth-fibre scattering into a compact neural appearance model. This motivates an adaptation to gemstones, where visible facets and their internal reflections are resolved in the image. Averaging their angular structure can remove the features the renderer must reproduce.

The research question is consequently: *how faithfully can a compact learned interior transport operator reproduce resolved gemstone appearance, and which operator errors explain its image-space failures?* The study begins with a radiance distribution map (RDM), diagnoses its missing spatial conditioning, and evaluates an operator that returns the outcome of an entire interior path. The aim is to test, rather than assume, whether predictive gains improve rendering.

The contributions are threefold. First, lookup and input-ablation experiments quantify the benefit of spatial conditioning for exit-triangle prediction. Second, a factorised operator supports stochastic and deterministic placement while preserving explicit geometric exit constraints. Third, an evaluation combining pixel statistics, geometric-facet averages, render-seed comparisons and path-component separation exposes a mismatch between operator accuracy and image agreement. Three cut families exhibit the pixel-level mismatch, while the step cut provides an informative counterexample at a coarser spatial scale.

The conclusions concern the evaluated models, budgets and scene, rather than all learned representations of specular media.

II. Background and Related Work

Aggregated appearance. The cloth model of Soh and Montazeri [1] learns an aggregate BSDF from traced fibre-level transport and includes an importance-sampling scheme. The present study adapts the idea of compressing internal scattering, not the cloth geometry or its complete implementation. The initial RDM is a directional summary. The proposed operator instead models a joint exit event conditioned on the entry state. Consequently, its results should be understood as an investigation of this adaptation, not a direct benchmark of the original cloth method.

Transport through bounded objects. Vicini et al. [2] learn shape-adaptive subsurface scattering using a conditional variational autoencoder and local shape descriptors. Their work establishes a relevant precedent for predicting transport through a region. Here the focus is a smooth dielectric with discrete reflection/refraction choices and sharply directed exits. Classical gemstone rendering explicitly models the relevant

optical mechanisms [3]; a spectral path tracer provides the numerical reference in this study [4], [5].

Conditional distributions and representations. Mixture density networks [6] represent multiple possible outcomes for an input. They are attractive when an entry state can escape by several internal routes. However, representing distinct outcomes does not by itself guarantee accurate branch probabilities or angular placement. Neural-network spectral bias [7] motivates investigating Fourier features [8]; local multiresolution hash encodings [9] provide another possible representation. The reported Fourier tests are limited experiments, not a test of every frequency encoding or network family.

Preserving variation and evaluating images. LEAN mapping [10] motivates retaining second moments rather than only means. The preliminary RDM experiment borrows this statistical idea, without reproducing LEAN's surface-filtering model. For image assessment, SSIM and FLIP [11], [12] offer complementary structural and perceptual criteria. The completed evaluation here instead emphasises linear-luminance agreement, highlight placement and facet averages with explicit render-noise baselines. No SSIM or FLIP measurements are claimed.

III. From a Radiance Map to an Interior Operator

The initial RDM bins exiting light by direction for each incoming direction. Its round-cut render lacks sharp internal structure. In 99.8% of well-populated bins, standard deviation exceeds the mean; the median ratio is 2.53. This identifies substantial within-bin variation, but not its placement.

A moment-based modification activates shading points with probability $p = 1/(1 + r^2)$ at gain $1/p$, where r is relative standard deviation. Position-keyed activation keeps the pattern fixed. This preserves the intended moments in expectation, but produces speckle rather than correctly located reflections: the missing position–direction relationship is not recovered.

Direction-conditioned tables predict exit triangles at accuracy 0.068 with 32 cells and 0.098 with 2,048, against 0.047 for the most frequent triangle. Conditioning on the entry triangle reaches 0.192. This motivates retaining position and normal alongside direction and wavelength; it does not exclude benefits from higher resolution in other representations.

IV. Method

A. Scene, geometry and reference transport

All principal comparisons use one studio scene: a perspective camera with a 30° field of view, a grey diffuse ground plane, textured area lights and two small intense sources. This is *one lighting rig with several sources*, not a single lamp. The non-uniform illumination makes exit-direction errors visible by assigning different radiance to different parts of the environment. Under spatially and directionally uniform incident radiance, many lossless transport differences would be much harder to observe.

Rendering uses the scalar spectral variant of Mitsuba 3 [13]. The reference samples Fresnel reflection or refraction, applies wavelength-dependent refraction using a diamond Sellmeier model, and traces each interior path until escape or a depth cap of 128. The estimator samples the BSDF without next-event estimation. Absorption and body colour are not modelled. The term *analytic reference* identifies these explicit transport rules; the finite-sample image is not asserted to be converged physical ground truth.

TABLE I
Geometry and primary-view evaluation support. Geometric facets are coplanar groups of mesh triangles. Retained facets cover at least 20 pixels.

Property	Round	Pear	Step
Vertices	34	33	58
Triangles / operator classes	64	62	112
Geometric facets	57	33	50
Retained visible facets	25	17	25
Stone-mask pixels	134,158	90,607	93,618
Training records	838,736	838,717	838,852
Validation records	209,656	209,712	209,696
Invalid records excluded	184	147	28

The three procedural meshes are a round brilliant, pear brilliant and step cut (Table I). The nominal round proportions use crown angle 34.5°, pavilion angle 40.75° and table fraction 0.56. Mesh validation checks closedness, manifold edges, outward orientation, positive signed volume and degenerate faces. The operator classifies *triangles*, which are not identical to optical facets: the 64-triangle round mesh contains 57 coplanar facet groups. This distinction is retained in the evaluation. Geometry hashes are stored with training data and checkpoints to detect unintended mesh mismatches; deliberate transfer is a separate experiment.

B. Conditional exit representation

The entry vector $\mathbf{x} \in \mathbb{R}^{10}$ comprises position $\mathbf{p}$, interior direction $\boldsymbol{\omega}$, entry normal $\mathbf{n}$ and wavelength λ. Positions are divided by the mesh radius and wavelength is encoded as $(\lambda - 595)/235$ with λ in nanometres. Let e denote escape before the depth cap, f an exit triangle and $\mathbf{y} \in \mathbb{R}^4$ exit coordinates. For escaping paths, the model factorisation is

$$p(e = 1, f, \mathbf{y} \mid \mathbf{x}) = s_\theta(\mathbf{x})\, \pi_\theta(f \mid \mathbf{x})\, q_\theta(\mathbf{y} \mid \mathbf{x}, f), \qquad (1)$$

with $p(e = 0 \mid \mathbf{x}) = 1 - s_\theta(\mathbf{x})$. The triangle and coordinate distributions are understood to be conditional on escape. This makes the non-escape event explicit instead of assigning an artificial exit to a truncated path.

Two coordinates encode barycentric logits. If triangle vertices are $\mathbf{v}_{f1}, \mathbf{v}_{f2}, \mathbf{v}_{f3}$, then

$$\boldsymbol{\beta} = \mathrm{softmax}(y_1, y_2, 0), \qquad \mathbf{p}_{\mathrm{out}} = \sum_{j=1}^{3} \beta_j \mathbf{v}_{fj}. \qquad (2)$$

The remaining two coordinates are outgoing tangent slopes:

$$\boldsymbol{\omega}_{\mathrm{out}} = \frac{\mathbf{n}_f + y_3 \mathbf{t}_f + y_4 \mathbf{b}_f}{\|\mathbf{n}_f + y_3 \mathbf{t}_f + y_4 \mathbf{b}_f\|}. \qquad (3)$$

These transformations place the exit on the selected triangle and in its outward hemisphere. They do not ensure that the sampled event corresponds to a physically valid internal route or that it cannot immediately hit another part of the stone.

C. Model variants and optimisation

A two-layer, width-64 SiLU encoder feeds escape and triangle heads. A learned 16-dimensional triangle embedding conditions the exit-coordinate decoder. The *stochastic* model uses four diagonal Gaussian mixture components in transformed coordinates [6]. The *clone* predicts a single coordinate vector with a smooth-L1 regression objective. The *deep clone* uses a four-layer coordinate decoder with a separate encoder. All variants still sample the exit triangle; "deterministic" refers only to placement given that triangle. The round-cut stochastic, clone and deep-clone models contain 17,637, 15,557 and 28,741 trainable parameters, respectively.

For escape label e_i, training combines binary escape loss with triangle and coordinate terms for escaping records:

$$\mathcal{L}_i = \mathrm{BCE}(s_\theta(\mathbf{x}_i), e_i) + e_i \left[-\log \pi_\theta(f_i \mid \mathbf{x}_i) + \ell_{\mathrm{coord}}(\mathbf{y}_i) \right]. \tag{4}$$

The stochastic coordinate term is mixture negative log-likelihood; the clone term is smooth-L1 error. Their total objective values are therefore not directly comparable. Adam [14] uses learning rate 10^{-3}, batch size 256 and gradient clipping at norm 10. Stochastic runs use 80 epochs and deep-clone runs 30; selection retains the lowest validation objective within each run.

These comparisons share the same training pool within a cut but change decoder family, objective, and in the deep case capacity and encoder arrangement. They are *model comparisons*, not a single-variable causal ablation. A separate same-checkpoint component-mean experiment isolates Gaussian sampling spread more directly.

D. Data and render integration

Each principal pool contains 1,048,576 path records from 65,536 entry states. Repeated outcomes capture alternative Fresnel decisions. Invalid records are excluded, and all outcomes of an entry state are assigned together to training or validation. Table I gives the resulting record counts. Validation selects checkpoints; separately recorded held-out tests are labelled as such rather than conflated with these validation scores.

Camera-matched entries initially cover an azimuth range but not every object orientation. Entry distribution is consequently a controlled part of the experiment, not a claim of complete surface coverage. Section VI-F examines the resulting pavilion blind spot.

During rendering, entry reflection or transmission is sampled using the analytic Fresnel rule. Transmitted entries are collected into batches of 512. Each sampled operator exit starts a new scene ray; non-escaping events terminate, and exits that immediately re-enter the stone are discarded. The outside scene and first-surface reflection use the same transport code in both modes. No importance correction makes the learned replacement an unbiased estimator of the original interior transport. It approximates that transport, and invalid-exit rejection can add further bias.

V. Evaluation Methodology

A. Sampling, masking and empirical noise baselines

Principal renders have resolution 512×512 and a box filter. Four independent seeds at $32\,\mathrm{spp}$ form a $128\,\mathrm{spp}$ average for each mode. Splitting the seeds into two groups yields independent $64\,\mathrm{spp}$ estimates. The main pixel comparison and reference self-comparison use these matching half-budgets; displayed images and facet means use the full $128\,\mathrm{spp}$ average. This distinction prevents a full-budget image score from being compared uncritically with a half-budget reference score.

Metrics use a fixed mask M of pixels whose central primary ray hits the stone. Linear RGB is converted to luminance with coefficients $(0.2126, 0.7152, 0.0722)$. No per-image brightness normalisation is applied before the reported energy measurements. Background and ground-plane caustics lie outside the mask. Thus, these results assess the visible stone, not the complete scene's transport.

Reference split-half agreement estimates sampling repeatability at that budget. It is not a strict upper bound on another method's correlation and does not establish convergence of the reference. Additional reference samples remain necessary to distinguish a stable fine-scale error from an unresolved rare highlight. Independent render seeds also do not replace independent training seeds [4].

B. Complementary image measurements

For neural luminance N_p and reference luminance A_p, the reported measures include Pearson correlation $\rho(N, A)$, signed relative energy error

$$\epsilon_E = \frac{\sum_{p \in M} N_p - \sum_{p \in M} A_p}{\sum_{p \in M} A_p}, \tag{5}$$

and relative root-mean-square error

$$\epsilon_R = \frac{\sqrt{|M|^{-1} \sum_{p \in M} (N_p - A_p)^2}}{\overline{A}_M}. \tag{6}$$

Highlight overlap is $|H_N \cap H_A| / |H_A|$, where each set contains the brightest 1% of masked pixels. Contrast is the luminance standard deviation divided by its mean. Correlation measures spatial association but can conceal brightness scaling. Energy error measures integrated brightness but not placement. Highlight overlap directly tests a small, visually important part of the image and is correspondingly sensitive to sampling noise. The metrics are interpreted jointly rather than combined into an arbitrary overall score.

C. Facet averages and component separation

To examine coarser structure, coplanar mesh triangles are grouped by normal and plane offset. Each visible group covering at least 20 pixels contributes its mean luminance; correlation and relative RMSE are computed across these

TABLE II
RECORDED ROUND-CUT EXIT-TRIANGLE BASELINES AND OPERATOR TESTS. NEAREST-NEIGHBOUR RESULTS USE 4,096 QUERIES OVER 1,046,359 STORED PATHS; THESE ARE HISTORICAL DIAGNOSTICS, SEPARATE FROM TABLE III.

Predictor	Top-1 accuracy
Most frequent triangle	0.047
Entry-direction table (32 cells)	0.068
Entry-direction table (2,048 cells)	0.098
Entry-triangle table	0.192
Nearest neighbour	0.287
Majority among eight neighbours	0.343
Stochastic operator	0.380
Clone operator	0.410

means, giving each retained facet equal weight. This spatial averaging reduces the influence of individual noisy pixels but also removes within-facet structure. It complements, rather than replaces, the pixel evaluation. Its equally weighted brightness statistics should not be interpreted as total image energy.

Each camera path is additionally tagged according to whether it passes through the interior replacement. The frame decomposes additively as $I = I_{\mathrm{operator}} + I_{\mathrm{untouched}}$. The reference uses the corresponding interior-path tag. This separates errors in approximated transport from direct reflections and scene contributions computed by unchanged code. High agreement in the untouched component is a useful control, but does not by itself validate the learned branch's integration.

D. Operator diagnostics and provenance

Operator evaluation measures exit-triangle top-1 accuracy and angular error of the decoded direction when supplied with the recorded true triangle. The latter isolates placement from triangle selection. Within-triangle *spread* around a prediction's own mean is a different quantity: a zero-spread prediction can still point in the wrong direction. Repeated transport outcomes also mean that triangle accuracy is not a complete assessment of distribution quality.

Tables and plots are assembled from saved evaluation JSON files and checkpoint validation histories. Historical diagnostics are identified separately from the matched three-cut render set. The companion source package records source paths and hashes. Training-loss curves and controlled timing experiments are not reconstructed from validation-only logs or undocumented measurements.

VI. RESULTS

A. Spatial conditioning and learned prediction

Table II shows improvement over the tested lookup baselines. The clone's 0.410 accuracy is approximately 43% above the nearest-neighbour score and 20% above the stronger eight-neighbour majority score. A lookup is a useful non-parametric comparator, not an upper bound on learnability. Differences in evaluation sample support should also be retained when interpreting rounded historical scores.

TABLE III
CHECKPOINT-SELECTION VALIDATION RESULTS FROM SAVED METRICS. S: STOCHASTIC; D: DEEP DETERMINISTIC. EPOCH IS THE SELECTED CHECKPOINT.

Cut	Model	Epoch	Triangle accuracy	Triangle CE
Round	S	80	0.371	2.547
	D	30	0.392	2.480
Pear	S	73	0.311	2.821
	D	29	0.338	2.736
Step	S	80	0.305	3.156
	D	30	0.328	3.067

In the recorded 60-epoch conditioning ablation, all arms use seed 23 and the same entry-grouped split. Removing position reduces top-1 accuracy from 0.404 to 0.273; keeping only direction yields 0.117. Removing wavelength changes it from 0.4042 to 0.4039. Position is therefore useful under this protocol, whereas wavelength contributes little to this particular triangle-classification score. The result does not demonstrate absence of dispersion in the physical system.

B. Validation behaviour

Figure 1 and Table III show that learning improves triangle prediction on all cuts. Deep models obtain higher selected validation accuracy and lower triangle cross-entropy than their stochastic counterparts. Several selected checkpoints occur at the final training epoch, including both round and step stochastic runs and their deep variants. Their curves therefore do not establish fully converged optimisation. This limits any claim that insufficient training or capacity has been eliminated.

For the round stochastic model, predicted mean escape probability is 0.9984 against a measured validation escape fraction of 0.9975. This is close agreement of *rates*, not proof of correct classification of individual escape events. Non-escape cases are rare, and an always-escape predictor already has high accuracy. The recorded Brier score is about 0.00249; discrimination of the rare class would require additional class-specific reporting.

C. Sharper images have lower pixel agreement

The main comparison is shown in Fig. 2 and Table IV. Under the tested rig, deterministic decoders produce sharper, higher-contrast images but lower pixel correlation and higher relative RMSE on every cut. Round deep-clone correlation is 0.038 versus 0.152 for the stochastic model; the corresponding pear values are 0.054 and 0.136, and step values are 0.168 and 0.293. The reference half-correlations are much higher, between 0.899 and 0.957.

The stochastic round image nearly preserves integrated energy, yet its contrast is only 0.93 against 3.78 for the reference. Energy agreement therefore does not imply recovery of structure. Conversely, the pear deep clone exceeds reference contrast (2.31 versus 1.48) and has a $+47.8\%$ energy error. A sharper-looking image is not necessarily an improvement in either placement or radiometry. Highlight overlap also remains

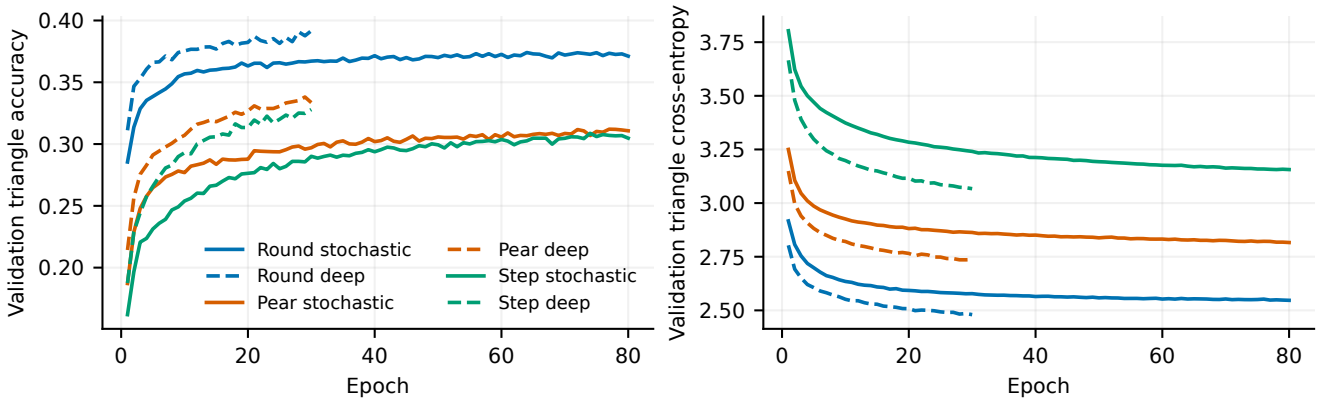


Fig. 1. Saved validation histories for the three cuts. Solid curves use the stochastic decoder; dashed curves use the deep deterministic decoder. Triangle accuracy and triangle cross-entropy are comparable across decoder families; total losses are not, because their coordinate objectives differ. These are single-training-seed histories, not mean curves with uncertainty bands.

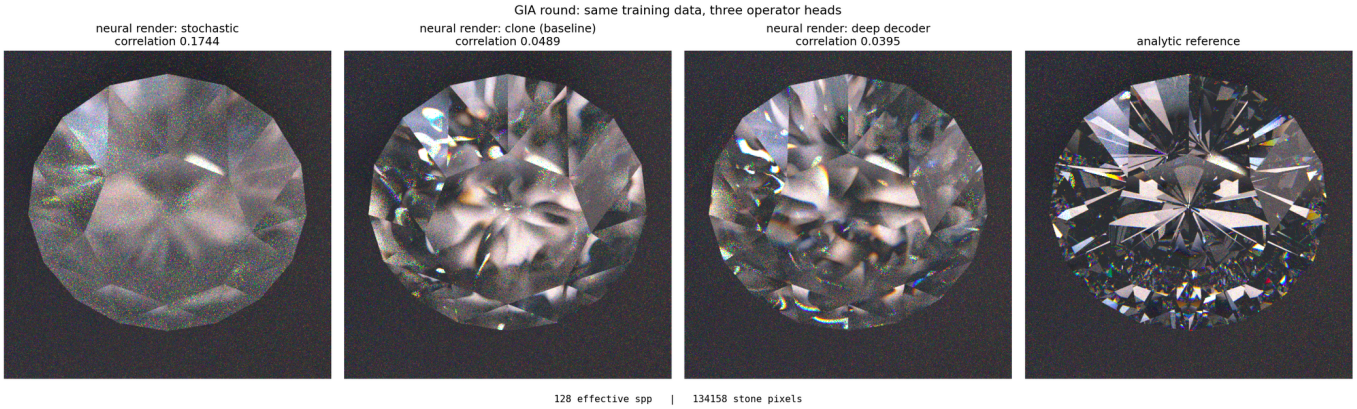


Fig. 2. Round-cut renders from the existing experiment: stochastic, clone, deep clone and analytic reference at 128 spp. Deterministic placement restores sharp features but does not reproduce their locations. Embedded panel scores use all render seeds; Table IV uses 64-spp half-splits.

well below reference self-overlap, independently supporting the observation of misplaced bright structure.

D. Coarse agreement is a distinct result

Facet averaging changes the interpretation (Fig. 4). Stochastic facet correlations are 0.399, 0.645 and 0.880 for round, pear and step cuts; deep-clone values are 0.038, 0.414 and 0.899. Reference facet self-correlations exceed 0.9995. Thus the step deep clone reproduces the ordering and magnitude of coarse facet brightness substantially better than its low pixel correlation alone suggests. Its facet relative RMSE is 0.333, compared with 0.376 for the stochastic model.

This is an important qualification of the central finding. Operator improvement does not invariably worsen *every* image statistic: the effect depends on spatial scale and the chosen quantity. Coarse agreement cannot establish correct sparkle or internal reflection placement, because a facet mean intentionally discards that structure. The combination of strong facet agreement and weak pixel agreement provides a more precise diagnosis than declaring the entire image either successful or unsuccessful.

E. Branch selection and within-branch placement

Historical held-out diagnostics distinguish spread from error. Within a recorded exit triangle, stochastic samples have a median angular spread of approximately $26.0°$, whereas the corresponding traced branches are almost collimated in the sampled experiment. A clone has no stochastic coordinate spread, but its mean direction error given the true triangle is still about $24.7°$. Determinism therefore removes one source of blur without establishing correct placement.

In a same-checkpoint intervention, sampling the mixture component but using its mean coordinates reduces measured spread from $26.0°$ to $8.1°$; the recorded image correlation falls from 0.174 to 0.153. The residual spread is consistent with retained discrete sampling. This intervention is more directly controlled than comparing separately trained decoder families.

Supplying the true exit triangle to the clone produces contrast 3.73 against 3.74 for its reference, but correlation remains 0.041 (Fig. 5). This implicates residual placement error even when triangle selection is bypassed. It does not establish that correcting triangle selection is generally harmful.

TABLE IV

MATCHED THREE-CUT IMAGE EVALUATION. EACH MODE HAS FOUR 32-SPP SEEDS (128 SPP TOTAL); CORRELATION, RELATIVE RMSE, HIGHLIGHT OVERLAP AND ENERGY ERROR USE 64-SPP HALF-IMAGES. CONTRAST USES THE FULL AVERAGE. "REFERENCE HALVES" COMPARES INDEPENDENT REFERENCE ESTIMATES AND IS AN EMPIRICAL SAMPLING BASELINE.

Cut	Decoder/comparison	Correlation ↑	Rel. RMSE ↓	Top-1% overlap ↑	Energy error	Contrast
Round	Stochastic	0.152	3.750	0.197	−0.8%	0.93
	Clone	0.048	4.741	0.071	+26.4%	2.35
	Deep clone	0.038	4.849	0.117	+13.8%	2.74
	Reference halves	0.957	1.140	0.777	+0.6%	3.78
Pear	Stochastic	0.136	1.834	0.007	+17.3%	0.89
	Deep clone	0.054	3.795	0.002	+47.8%	2.31
	Reference halves	0.899	0.684	0.595	+0.4%	1.48
Step	Stochastic	0.293	2.353	0.059	+14.2%	0.84
	Deep clone	0.168	2.956	0.057	+15.3%	1.77
	Reference halves	0.942	0.817	0.650	−0.3%	2.37

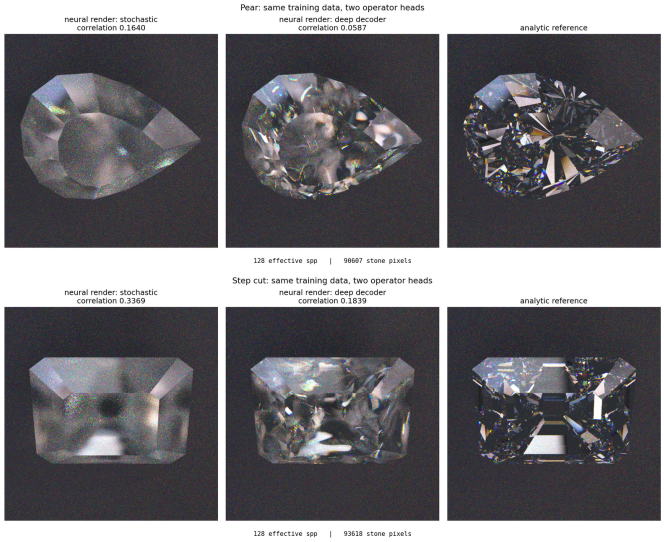


Fig. 3. Pear (top) and step (bottom): stochastic and deep-clone results beside their references. Broad shape and brightness organisation can survive even when fine internal reflections are incorrect. Images use 128 spp.

F. Data, capacity, encoding and coverage

Recorded data-ladder accuracy rises from 0.048 at 256 entry states to 0.332 at 16,384 and 0.347 at 65,536. This shows diminishing gains over the tested range, not saturation under arbitrary data growth. A wider stochastic model improves validation objective by about 0.33 nats before overfitting; a deeper decoder reduces recorded placement error from 24.5° to 21.6°. Three tested Fourier scales yield errors of 25.8°–26.7° against the 21.6° control. These interventions do not remove the observed failure. Unequal optimisation progress, a limited capacity range and single training seeds prevent them from excluding those factors completely.

Nearest-neighbour placement is about 8.9° *on queries where its chosen triangle is correct*. This subset differs from evaluating a learned decoder with the true triangle supplied for every query. The value is consequently a diagnostic of locally recoverable exits, not a directly comparable universal placement bound. Likewise, differing exits among nearby entries may reflect both branch-boundary sensitivity and Fresnel branching; neighbourhood diversity alone cannot prove an unsampleable discontinuity density.

Camera-matched training initially supplies no pavilion entries. In a recorded 60-frame pear tumble, pavilion-facing poses reduce predicted escape to 46%. Adding uniformly directed entries with a disc aim offset places approximately 53% of entries on the pavilion and restores recorded escape to 100% across the sequence. At a pavilion-facing pose, correlation remains 0.018 against reference repeatability of 0.951. Coverage repairs a specific failure mode but does not by itself recover placement. The tumble is a coverage diagnostic, not a substitute for a held-out lighting and viewpoint benchmark.

G. Transfer and apparent agreement from untouched light

The categorical head is mesh-specific. A separate experiment maps a pear-trained operator to an unseen round variant by matching facet normals. This changes geometry and correspondence simultaneously, so it tests a particular retargeting heuristic rather than geometry-independent generalisation.

Whole-stone correlations of approximately 0.37 and 0.35 initially suggest improved agreement. However, about 90% of a dominant crown highlight is untouched light. Removing an 81×81 patch around it reduces the recorded correlation to -0.001, and the operator component correlates at only 0.004. On the default round experiment, analogous patch removal leaves a recorded correlation of 0.205 essentially unchanged. The lesson is methodological: agreement produced by a shared analytic reflection is not evidence that the learned interior transport transfers.

H. Recorded computational cost

Table V reports the median of the saved per-seed durations, not the time to obtain the complete four-seed average. None of the recorded neural configurations is faster than its cut's reference in these logs. Their interpretation is limited by uncontrolled system load and incomplete hardware provenance. Earlier project notes report 5–9 μs per batched query,

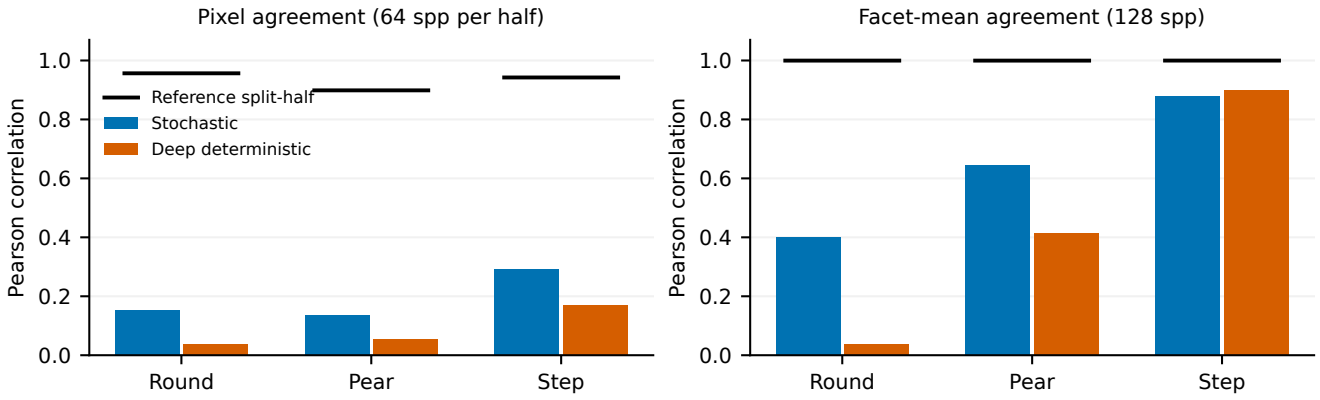


Fig. 4. Image agreement at two spatial scales. Left: pixel half-split correlation. Right: correlation of full-seed facet means, with each retained facet weighted equally. Black marks indicate reference split-half agreement at the corresponding scale. The step deep clone improves coarse facet agreement while worsening pixel agreement. The different estimands and sample budgets must not be interpreted as a pure denoising comparison.

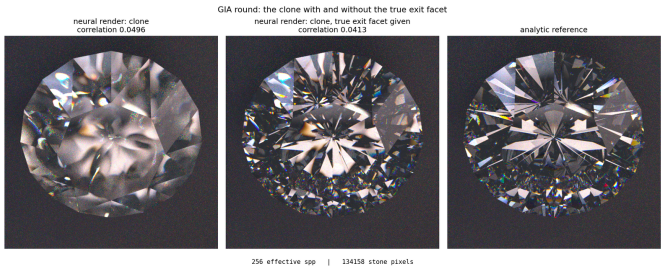


Fig. 5. Historical 256-spp diagnostic: clone, clone supplied with the traced exit triangle, and reference. Supplying the correct triangle restores contrast but does not correct the learned exit coordinates.

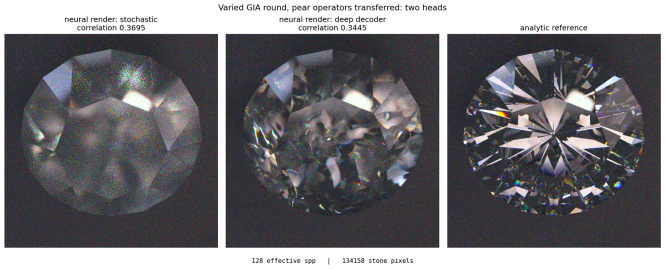


Fig. 6. Retargeting diagnostic from the recorded experiment. A direct lamp reflection on the crown contributes apparent agreement although it bypasses the learned operator. Component-separated evaluation exposes this effect.

but that figure is insufficient for an end-to-end speed claim: input preparation, sampling, decoding and scene-ray bookkeeping also contribute. No controlled training-time, memory or throughput benchmark is asserted here.

VII. Discussion

A. Why predictive gains need not improve the image

The image depends on the exit distribution weighted by the radiance encountered after escape. Schematically, for entry state $\mathbf{x}$ and exit state z,

$$L(\mathbf{x}) = \int L_{\text{ext}}(z)\, p(z \mid \mathbf{x})\, \mathrm{d}z. \tag{7}$$

TABLE V
Median recorded seconds per 512×512, 32-spp render, over four seeds. These historical runs were not a controlled hardware benchmark; cross-run differences may include concurrent load.

Mode	Round	Pear	Step
Analytic	657	485	497
Stochastic	1,269	955	746
Deep clone	767	725	655

A training objective that weights all recorded exits similarly need not prioritise the exits that connect to narrow bright sources. A small angular error near a source boundary can have a large image effect, whereas a larger error between equally dark directions may contribute little. Triangle accuracy and average angular error are therefore useful but incomplete proxies for rendered fidelity.

A broad predicted distribution can spread energy over both correct and incorrect regions, preserving some correlation while suppressing contrast. A sharper but misplaced distribution can lose this overlap. The results are consistent with this explanation, but do not establish a universal monotonic relationship between blur and correlation. The facet-scale step result shows explicitly that an improvement can be visible in one image statistic while remaining absent in another.

The representation is one plausible contributor. Smooth conditional densities approximate transport that contains sharply concentrated branches, and changes of internal route may create rapidly varying exit behaviour. Spectral bias provides a relevant motivation [7], but incomplete convergence and the tested objectives remain alternative or interacting explanations. Local encodings, route-aware representations and radiance-weighted objectives are hypotheses for further work, not demonstrated solutions.

B. What the current evidence supports

The evidence supports three bounded conclusions. Spatial conditioning improves exit-triangle prediction in the tested setting. Improved validation prediction does not guarantee improved fine-scale image agreement. Finally, reporting pixel, facet and operator-only measurements together yields a more informative assessment than a single whole-image score. The study provides an empirical failure analysis of a compact learned replacement, rather than a proof that neural transport cannot represent specular gemstones.

VIII. Limitations and Further Validation

The study uses one rig, mostly one pose, simplified meshes and mesh-specific categorical heads. Each principal configuration has one training seed; several validation curves are still improving. Reference convergence remains unverified, stone masking excludes ground caustics, and saved logs lack paired training-loss curves. Historical timings are not controlled performance measurements.

Four proposed tests address these gaps: (1) return traced exits through the learned operator's interface to measure integration error and rejection; (2) repeat training with at least three seeds and adequate optimisation; (3) test reference stability at 128, 512 and 2,048 spp, extending if necessary; and (4) evaluate fixed models under broad and narrow-source lighting at several views inside training coverage. These are future experiments, not reported results.

For a more noise-tolerant supplementary evaluation, normalised mean absolute error can be reported as

$$\mathrm{NMAE}(N, A) = \frac{\sum_{p \in M} |N_p - A_p|}{\sum_{p \in M} A_p + \epsilon}, \tag{8}$$

where ϵ is a documented small denominator safeguard. Evaluating it after matched 2×2, 4×4 and 8×8 block averaging would separate spatial scales; native-resolution scores must remain visible so that averaging does not hide the failure being studied. These NMAE and multiscale scores have not yet been measured. Their uncertainty should be assessed with independent render seeds, alongside reference self-comparisons, rather than assuming any metric is immune to Monte Carlo noise.

A controlled performance study should record hardware, software, threads, precision, warm-up, batching and model-loading policy. It should separate network latency, complete operator-call latency and render time, and compare image error at equal elapsed time. Mesh and checkpoint hashes, scene parameters and exposure settings are also needed for reproducibility.

IX. Conclusion

A direction-conditioned radiance map was generalised to a position-conditioned interior transport operator for three procedural gemstone cuts. The learned models improve exit-triangle prediction over the tested lookup baselines, but the deterministic variants' higher validation accuracy does not translate into better fine-scale image agreement under the evaluated studio rig. Highlight overlap and relative RMSE support the correlation-based finding, while facet-averaged evaluation reveals substantial coarse agreement on the step cut. True-triangle and component-separation diagnostics further distinguish placement error from branch selection and from apparent agreement due to untouched reflections. The resulting contribution is a measured account of where this approximation succeeds and fails. Stronger claims about representation limits, generalisation or computational advantage require the additional convergence, repeatability and performance controls identified here.

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